

General Purpose Technology

A pervasive technology that improves over time and spawns complementary innovation

GLOSSARY 05 - SHEET 2 / 8

PROFESSIONAL DEFINITION

A general purpose technology is a technology characterised by three properties: pervasiveness across many sectors, continuing improvement over time that lowers the cost of use, and the capacity to spawn complementary innovation in the sectors that adopt it. Steam power, electricity and the semiconductor are the canonical cases; the economic significance of such technologies lies less in their direct output than in the innovation they enable elsewhere (Bresnahan and Trajtenberg, 1995).

WHO DEFINED IT ORIGINALLY

Timothy F. Bresnahan and Manuel Trajtenberg defined the category in the Journal of Econometrics in 1995. Elhanan Helpman (1998) assembled the growth-theoretic treatment; Paul A. David's (1990) study of the dynamo established the long delay between availability and productivity effect. Application to AI is developed by Agrawal, Gans and Goldfarb (2018) and Brynjolfsson, Rock and Syverson (2021).

WHY THIS DEFINITION WAS DEVELOPED

Growth accounting treated technologies as separable inputs, which could not explain why a small number of innovations produced economy-wide transformation while most did not. The general purpose category identifies the structural properties that generate innovational complementarities, and thereby explains the characteristic delay: the enabling technology must be widely deployed before the complementary innovations that carry the productivity effect become possible.

FIGURE 1 - THREE DEFINING PROPERTIES

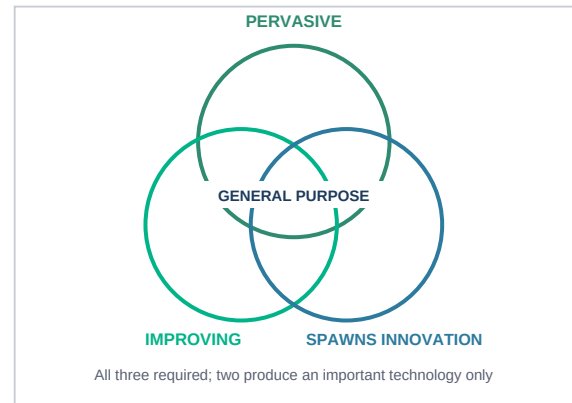


Figure 1. Many technologies satisfy two of the three properties. The third - the capacity to raise the return to innovation in adopting sectors - is what produces economy-wide effect, and it is also the source of the characteristic multi-decade delay (after Bresnahan and Trajtenberg, 1995).

HOW IT IS USED

Test the three properties against the technology in question, then plan on the historical delay pattern rather than on vendor timelines; the value accrues to complementary innovation in the adopting sector, not to the enabling technology itself.

WHEN IT IS USED

In long-horizon strategic planning, in policy analysis of AI's economic effects, when assessing whether an AI investment should be treated as infrastructure or as an application, and in board discussion of pace.

BY WHOM IT IS USED

Corporate strategists on long horizons, economists and policy analysts, investors assessing structural rather than cyclical change, and executives calibrating expectations about timing.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Its predecessor in the growth literature was Schumpeter's account of innovation clusters, which described bunching in time without specifying the mechanism. Bresnahan and Trajtenberg supply the mechanism as innovational complementarity: the enabling technology raises the return to innovation in the adopting sector, which raises the return to further improvement in the enabling technology. David's dynamo evidence then converts the framework into a timing prior - electrification took some four decades to appear in productivity statistics.

RELATED DEFINITIONS IN THIS COURSE

Artificial Intelligence - Unit 1

Productivity Paradox - Unit 5

Disruptive Innovation - Unit 1

Skill-Biased Technological Change - Unit 5

Digital Divide - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

Agrawal, A., Gans, J. and Goldfarb, A. (2018) Prediction Machines: The Simple Economics of Artificial Intelligence. Boston, MA: Harvard Business Review Press.

Bresnahan, T.F. and Trajtenberg, M. (1995) 'General purpose technologies: engines of growth?', Journal of Econometrics, 65(1), pp. 83-108.

David, P.A. (1990) 'The dynamo and the computer: an historical perspective on the modern productivity paradox', American Economic Review, 80(2), pp. 355-361.

Helpman, E. (ed.) (1998) General Purpose Technologies and Economic Growth. Cambridge, MA: MIT Press.